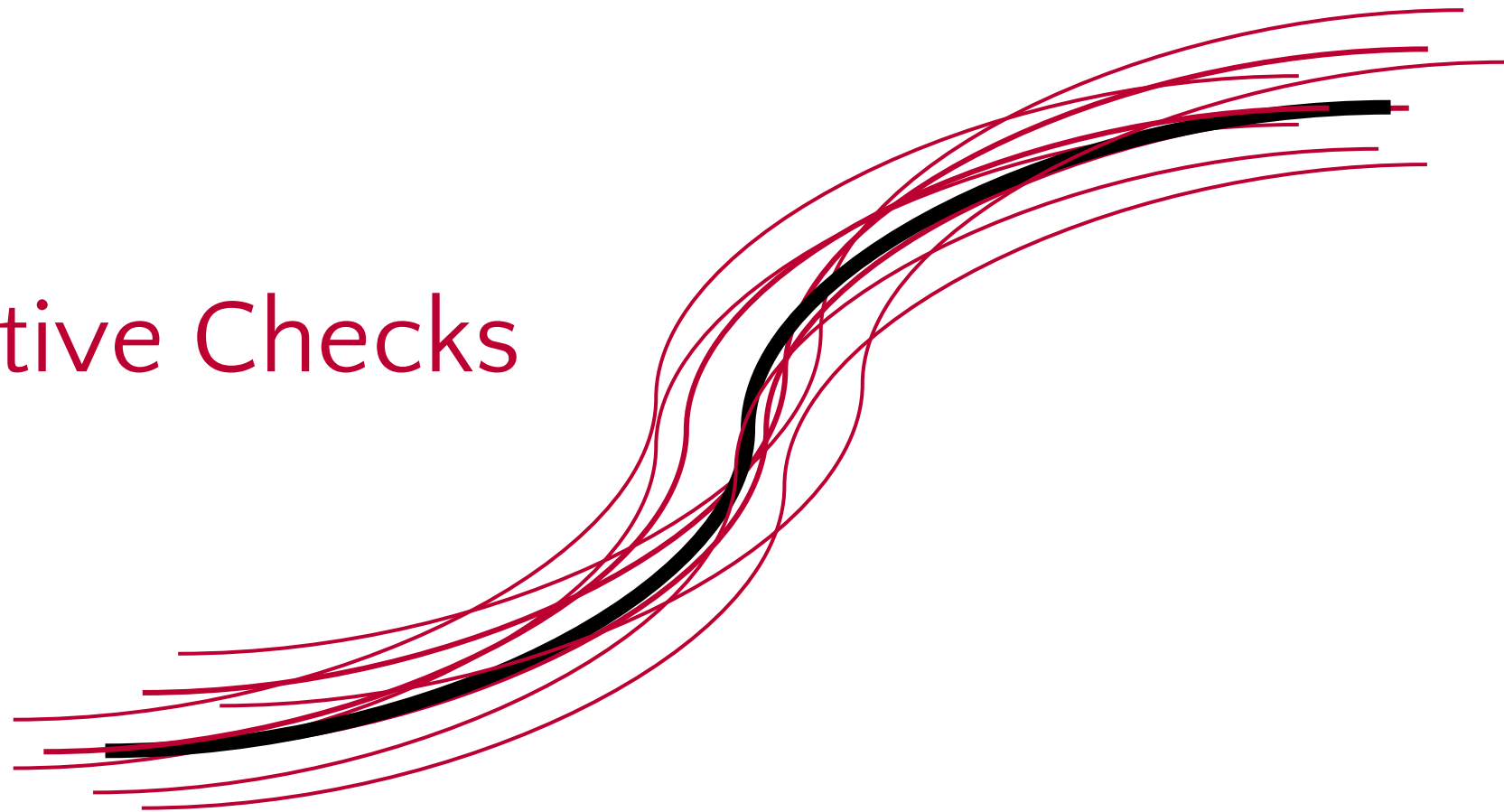


Prior Predictive Checks



Simple vs. Complex Prior Structures



$$p(\theta) = \text{Beta}(\theta \mid a, b)$$



Simple vs. Complex Prior Structures



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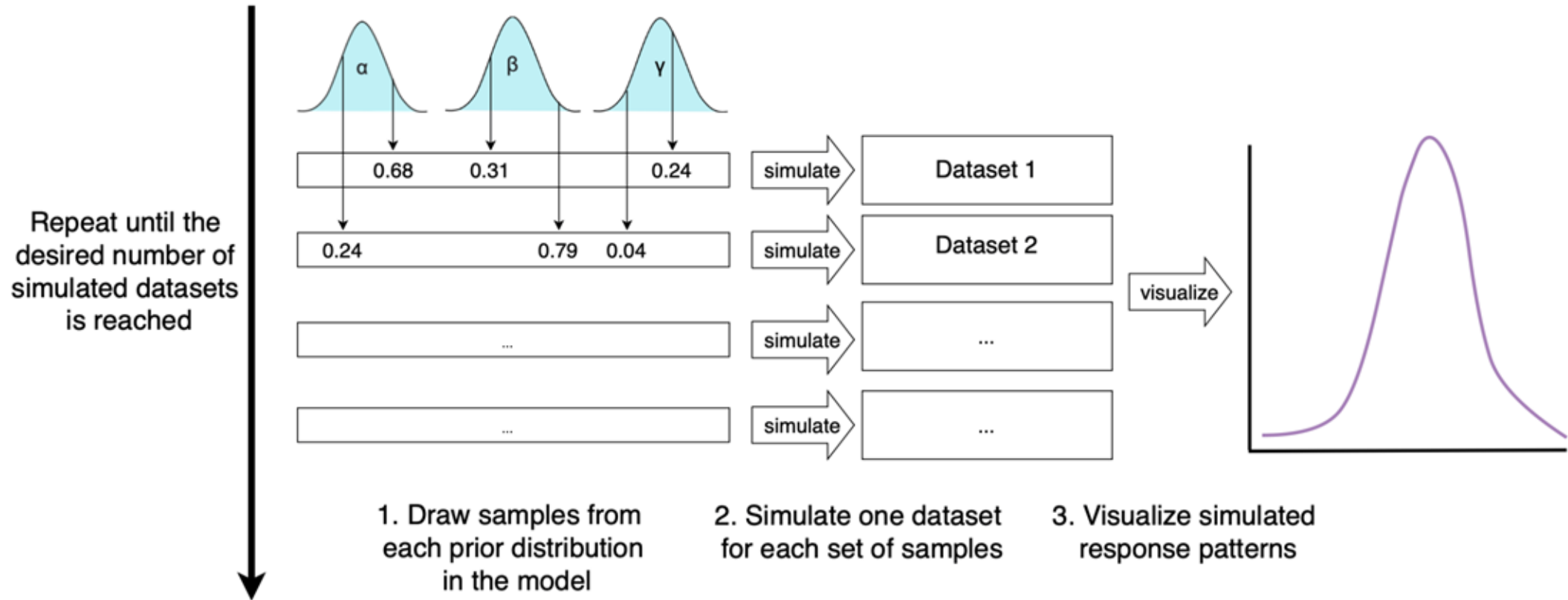


$$\mu_{i,j,k} = \begin{cases} \frac{c_{i,j,k} - \beta_{i,j}}{\alpha_{i,j}} & \text{if } c_{i,j,k} \geq \beta_{i,j} \\ 0 & \text{if } c_{i,j,k} < \beta_{i,j} \end{cases}$$

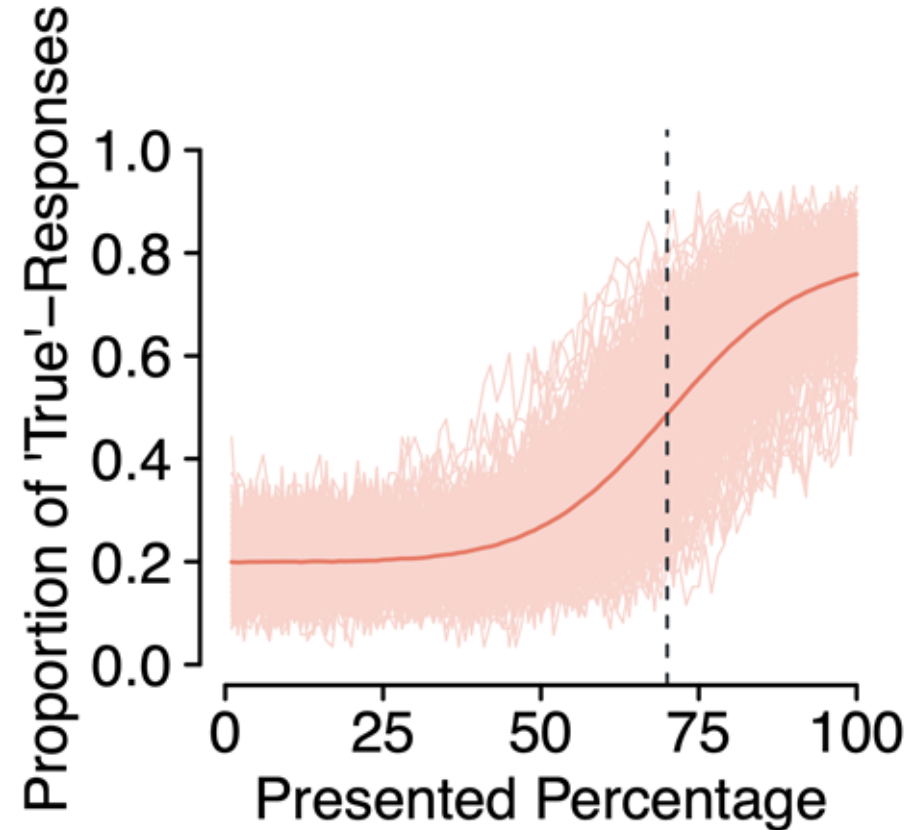
$$Y_{i,j,k} \sim \text{Bernoulli}(\pi_{i,j,k}),$$

$$\pi_{i,j,k} = \gamma_{i,j} + (1 - 2\gamma_{i,j}) \times (1 - \exp(-\mu_{i,j,k}))$$

Prior Predictive Checks



Prior Predictive Checks



$$\mu_{i,j,k} = \begin{cases} \frac{c_{i,j,k} - \beta_{i,j}}{\alpha_{i,j}} & \text{if } c_{i,j,k} \geq \beta_{i,j} \\ 0 & \text{if } c_{i,j,k} < \beta_{i,j} \end{cases}$$

$$Y_{i,j,k} \sim \text{Bernoulli}(\pi_{i,j,k}),$$

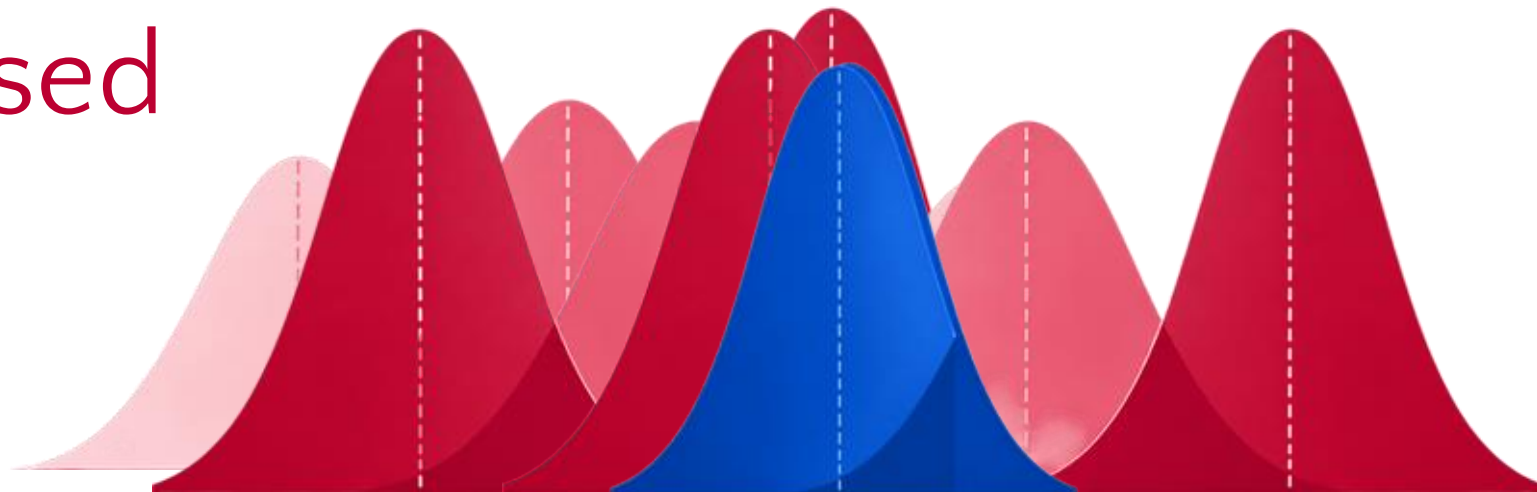
$$\pi_{i,j,k} = \gamma_{i,j} + (1 - 2\gamma_{i,j}) \times (1 - \exp(-\mu_{i,j,k}))$$



Prior Predictive Checks

Let's try this out with **brms**.

Simulation-based Calibration



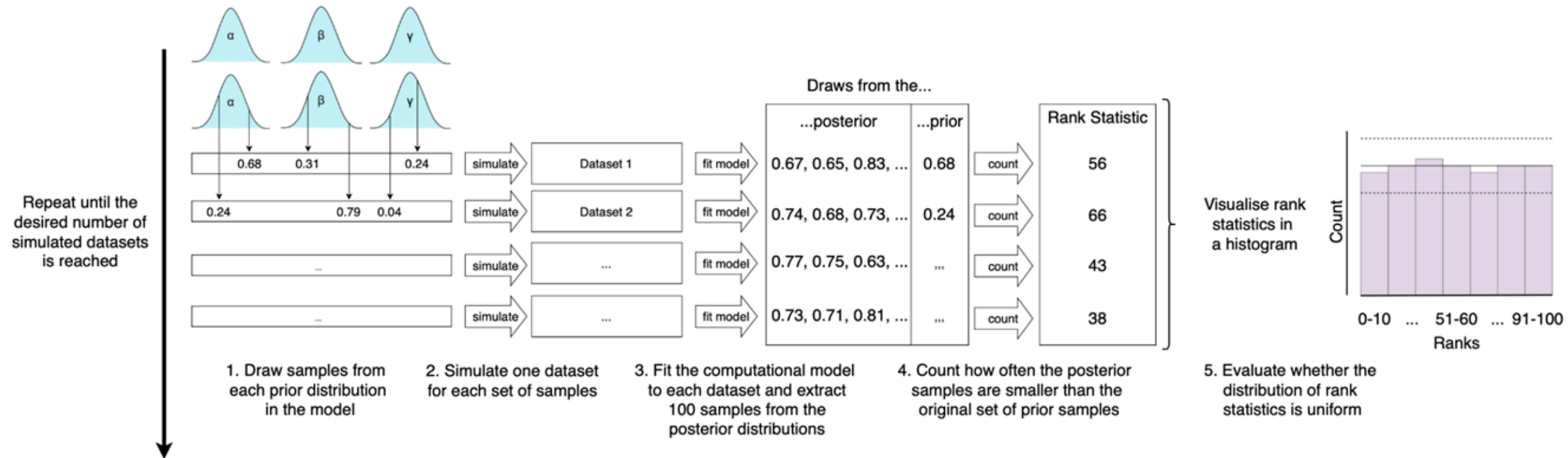
Simulation-based Calibration



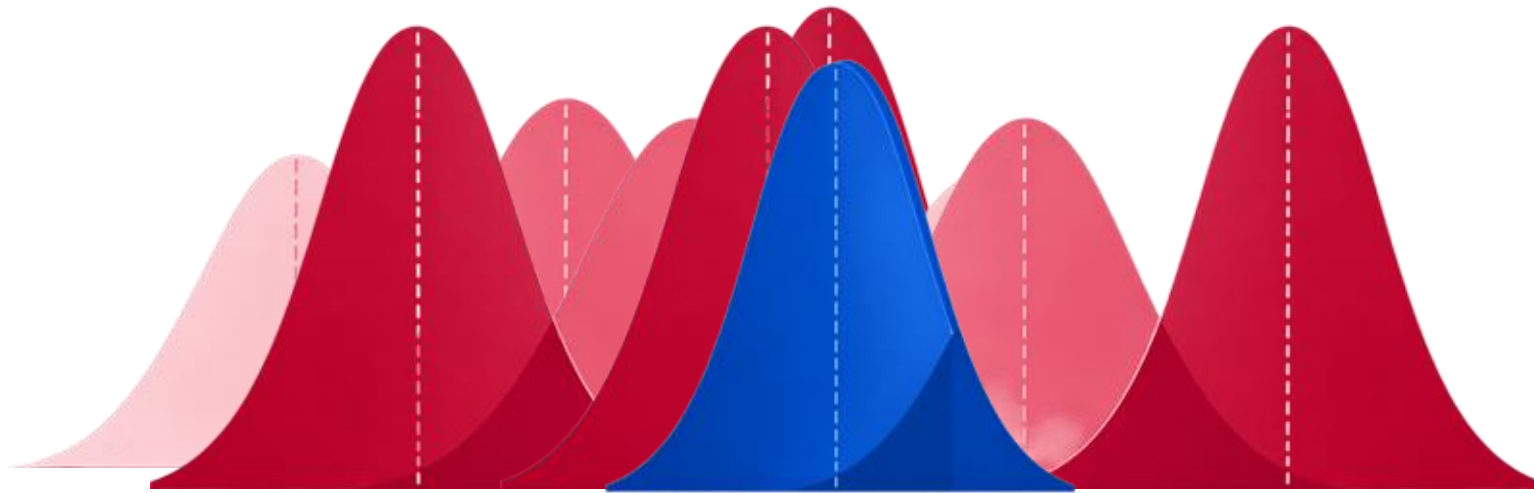
```
data {  
  // int<lower=1> D;           // # Dimensions of the model: Nr. of quantifiers  
  int<lower=1> I;             // # Participants  
  int<lower=0> N;             // # Observations/Rows in dataset  
  int<lower=0,upper=1> y[N];  // # Data 0,1  
  vector[N] cperc;           // # Centered Percentages  
  int<lower=1,upper=I> sub[N]; // # Participant vector  
  real mu_b_mean;  
}  
  
parameters {  
  vector<lower=-6, upper=0>[I] log_a;  
  vector<lower=-0.5, upper=0.5>[I] b;  
  vector<lower=0, upper=0.5>[I] g;  
  real<lower=-6, upper=0> mu_log_a;  
  real<lower=0, upper=0.5> mu_g;  
  real<lower=-0.5, upper=0.5> mu_b;  
  real<lower=0> sigma_mu_log_a;  
  real<lower=0> sigma_mu_g;  
  real<lower=0> sigma_mu_b;  
}  
  
transformed parameters {  
  vector<lower=0, upper=1>[I] a;  
  for (i in 1:I) a[i] = exp(log_a[i]);  
}  
  
model {  
  vector[N] mu;  
  vector[N] p;  
  // individual level parameters  
  for (i in 1:I) log_a[i] ~ normal(mu_log_a, sigma_mu_log_a);  
  for (i in 1:I) g[i] ~ normal(mu_g, sigma_mu_g);  
  for (i in 1:I) b[i] ~ normal(mu_b, sigma_mu_b);  
  
  // hyperparameters: means and standard deviations of multivariate distributions  
  mu_log_a ~ normal(-3, 1);  
  mu_b ~ normal(mu_b_mean, 0.1);  
  mu_g ~ normal(0.17, 0.05);  
  
  sigma_mu_log_a ~ inv_gamma(2, 1);  
  sigma_mu_g ~ inv_gamma(12, 1);  
  sigma_mu_b ~ inv_gamma(8, 1);  
  
  for (n in 1:N) mu[n] = (cperc[n] - b[sub[n]]) / a[sub[n]];  
  for (n in 1:N) p[n] = g[sub[n]] + (1 - 2 * g[sub[n]]) * (1 - exp(-fmax(mu[n], .001)));  
  y ~ bernoulli(p);  
}  
  
generated quantities {  
  vector[N] mu_pred;  
  vector[N] p_pred;  
  int y_pred[N];  
  // Prior Predictive  
  for (n in 1:N) mu_pred[n] = (cperc[n] - b[sub[n]]) / a[sub[n]];  
  for (n in 1:N) p_pred[n] = g[sub[n]] + (1 - 2 * g[sub[n]]) * (1 - exp(-fmax(mu_pred[n], .001)));  
  y_pred = bernoulli_rng(p_pred);  
}
```

Godmann, H. R., Haaf, J. M., Visser, I., Szymanik, J., & Sarafoglou, A. (in press). Toward a principled Bayesian workflow in semantics and pragmatics: A tutorial. Cambridge University Press.

Simulation-based Calibration

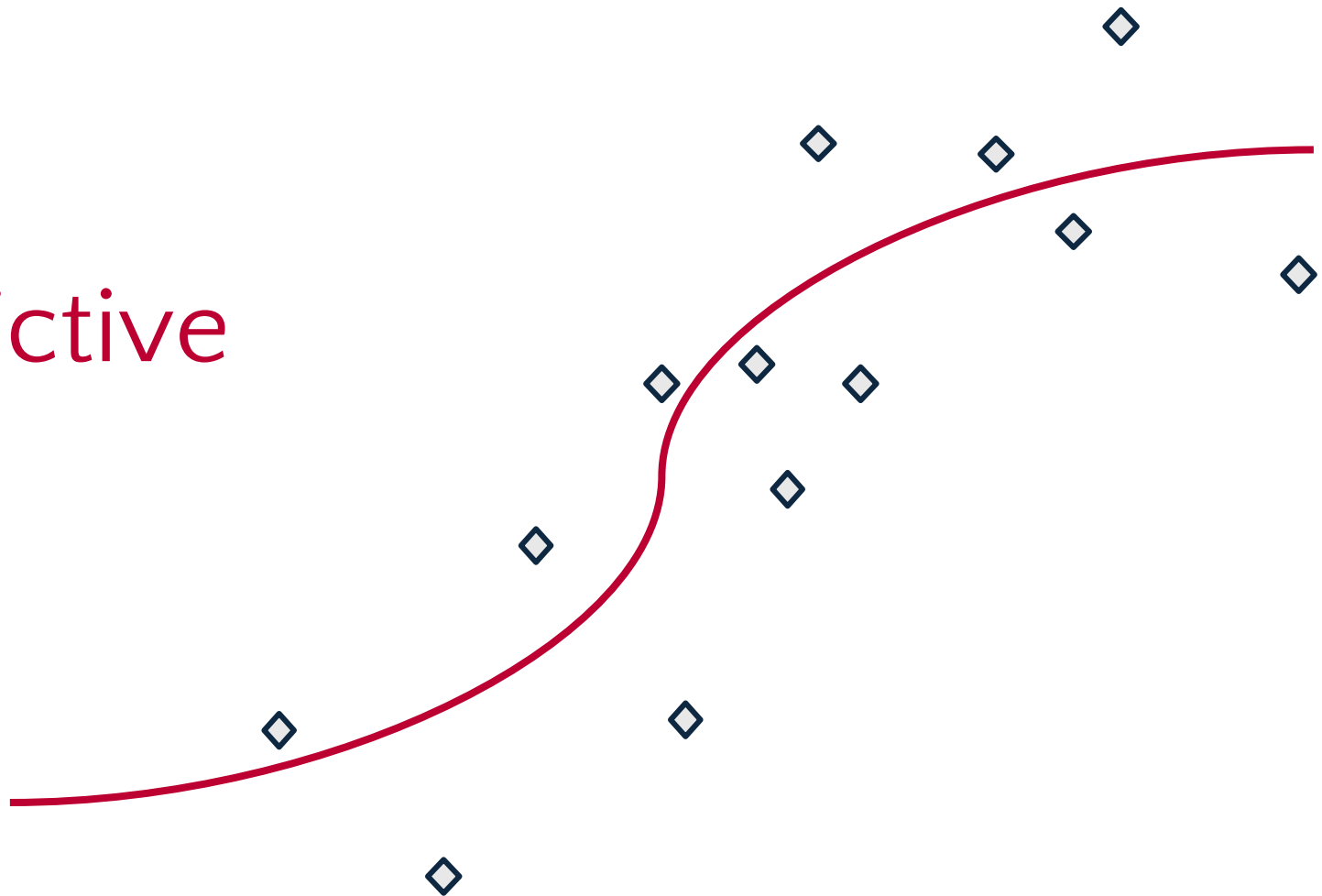


Simulation-based Calibration

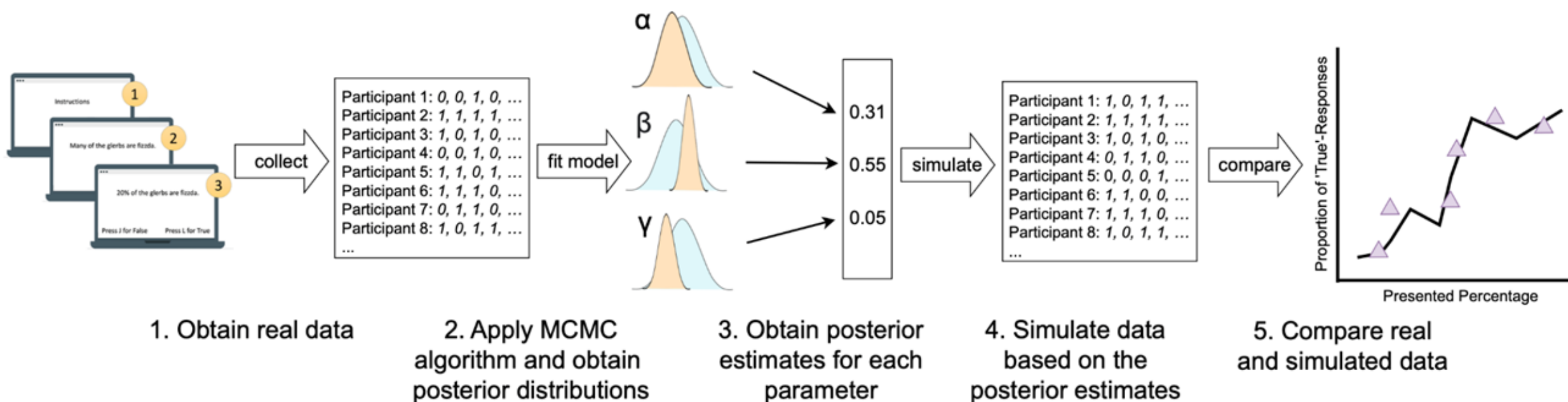


- Prior distribution (base for many data sets)
- Posterior distributions (fits to all these data sets)

Posterior Predictive Checks



Posterior Predictive Checks





Posterior Predictive Checks

Let's try this out with **brms**.

Bayesian Workflow – Further Reading

Godmann, H. R., Haaf, J. M., Visser, I., Szymanik, J., & Sarafoglou, A. (in press). Toward a principled Bayesian workflow in semantics and pragmatics: A tutorial. Cambridge University Press.

Schad, D. J., Betancourt, M., & Vasishth, S. (2021). Toward a principled Bayesian workflow in cognitive science. *Psychological methods*, 26(1), 103.

Talts, S., Betancourt, M., Simpson, D., Vehtari, A., & Gelman, A. (2018). Validating Bayesian inference algorithms with simulation-based calibration. arXiv preprint arXiv:1804.06788.